Class 19 Options

Options vs Futures

Pricing options,

the one period problem

Replicating portfolio

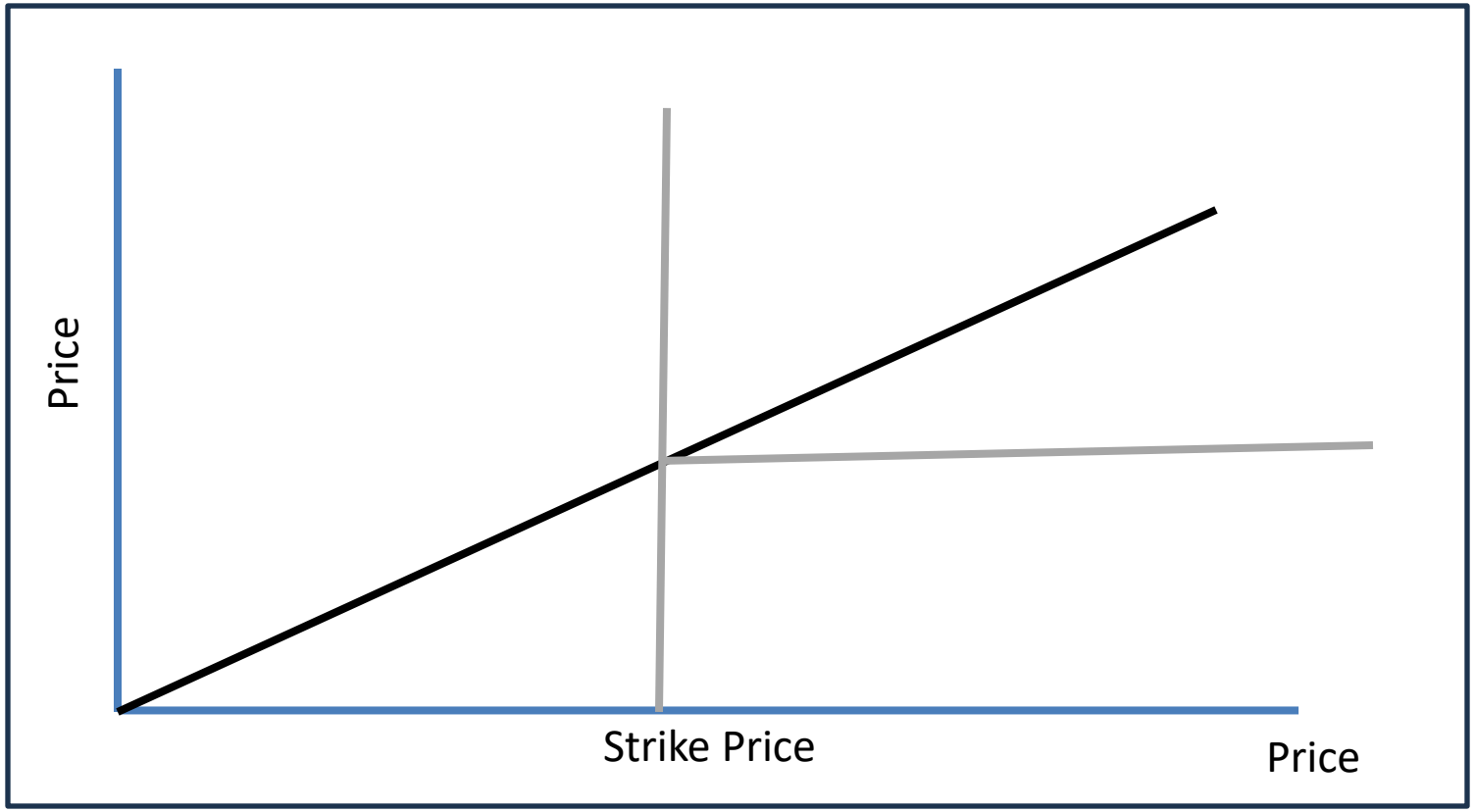
Key concepts

- Call vs put option
- Volatility and the value of the options
- Expiry date
- American vs European option
- Replicating portfolio

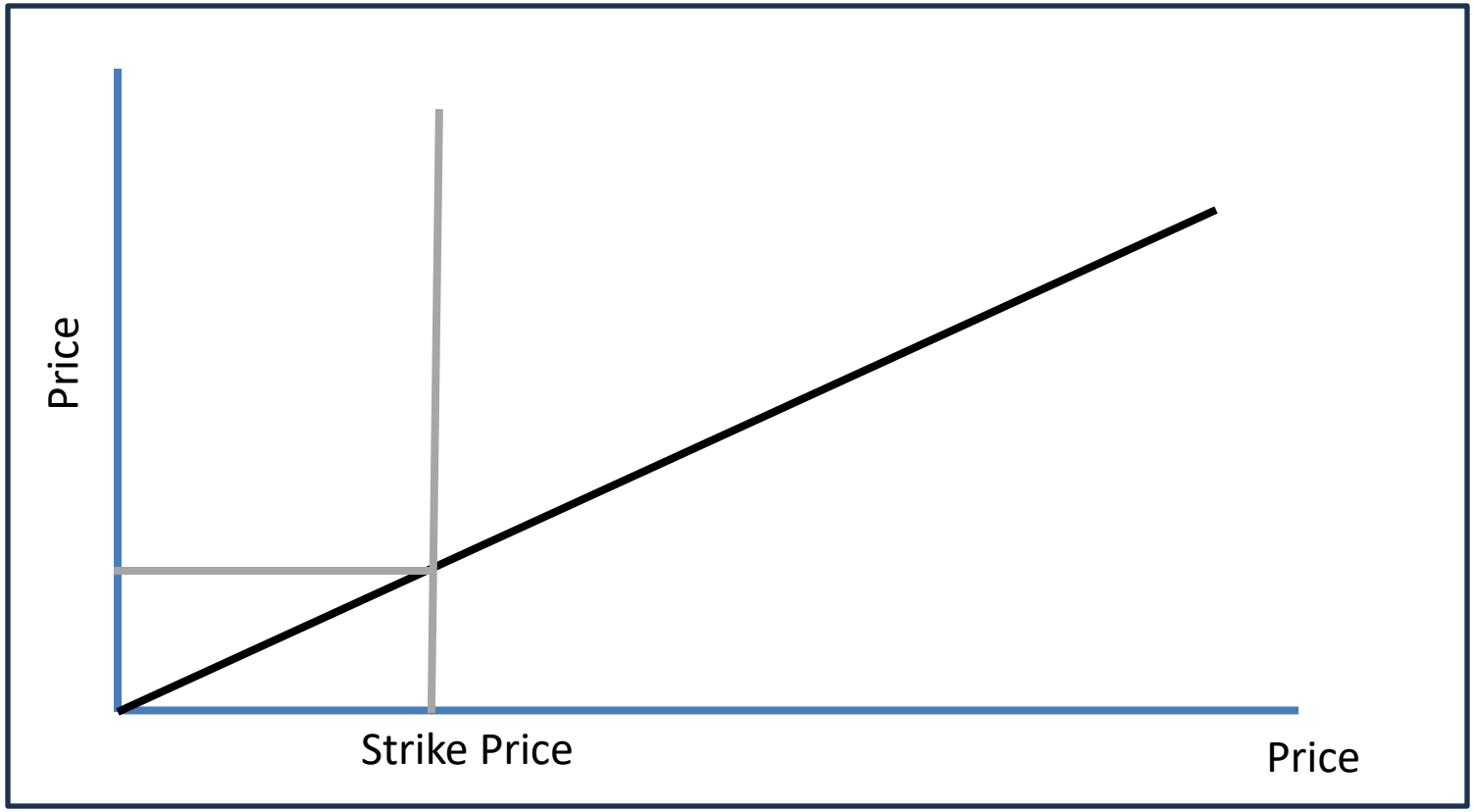
Options

- **Definition**
- **Call Option:** The right to buy an item at a pre-specified price up to some date.
  - Example the right to buy 100 shares of HSBC common stock for $55 on or before 5 pm March 13 2026.
- **Put Option:** The right to sell an item at a pre-specified price up to some date.
  - Example the right to sell 100 shares of HSBC common stock for $55 on or before 5 pm March 13 2026.
- $55 is the strike price
- A call is a right to buy a put is a right to sell
  - Call becomes valuable if price rises
  - Put if price falls

Call option as a picture



Put option as a picture



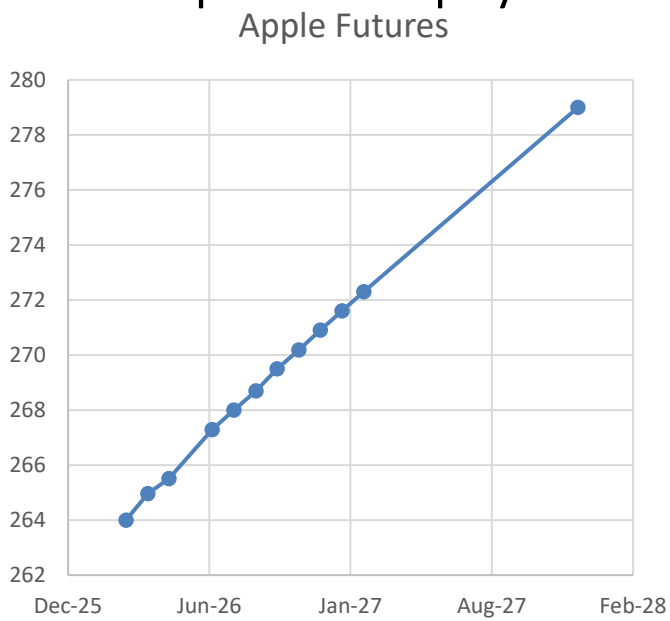
Futures vs Options

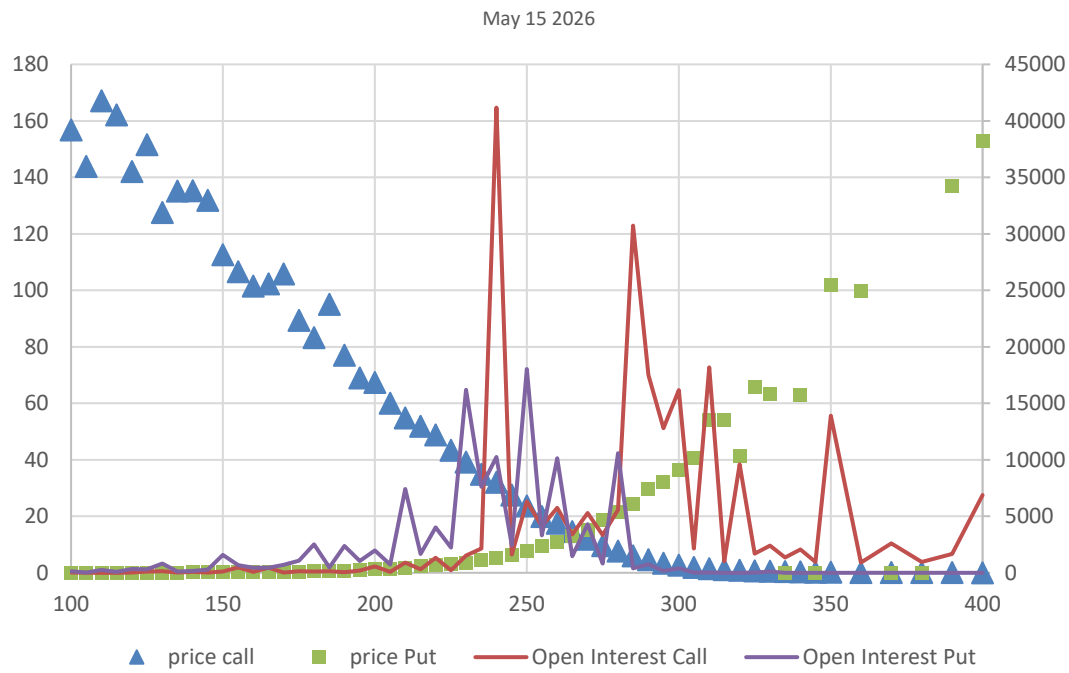
- Like futures, options are contracts where things happen in the future
- Unlike futures they are not firm on both sides (one side decides).
- This is not unusual, options embedded in many contracts
  - Mortgages
  - Credit cards
  - Coupons

Apple
Underlying: AApl 1 price 264.58

- Futures
  - Eurex futures 12 contracts available
  - Depend on expiry date

- Options
- Monthly December 28
- Weekly end of April





https://www.eurex.com/ex-en/markets/equ/fut/Apple-2408726

Options expiring May 15 2026

Calls				Puts		
Price	Open Int	Imp Vol	Strike price	Price	Open Int	Imp Vol
32	41,173	31%	240	5.35	10,249	31%
27.5	1,609	30%	245	6.35	2,442	30%
23.68	6,315	29%	250	7.6	18,033	29%
19.9	4,138	28%	255	9.5	3,307	29%
17.6	5,741	28%	260	10.85	10,142	28%
14.65	3,406	27%	265	12.85	1,462	27%
11.84	5,294	26%	270	15.2	4,291	26%
9.65	3,330	26%	275	18.53	806	26%
7.65	5,584	25%	280	21.55	10,582	25%
5.95	30,715	25%	285	24.3	400	25%

Option types

- **American options**: you can exercise them whenever you want after the contract is signed.
  - The right to buy 100 shares of JPMorgan common stock for $130 anytime before April 21 at 5 pm.
- **European options:** you can only exercise them at the expiry date
  - The right to buy 100 shares of JPMorgan common stock for $130 on April 21 at 5 pm.
- **Option is defined by:** the asset you have contracted over:(100 shares of JPMorgan common stock); The strike price: $130; Expiration date: April 21 2026, 5 pm; Type: European or American, and Put vs Call.
- Like other financial assets there some publicly traded options (see Chicago Board of Trade), and some are over the counter.

Why options?

- Why not futures?
- Consider the case of an airline
- On the revenue side It sells tickets up to a year in advance.
- On the cost side
  - it does not know what fuel prices will be (it does know its other costs) and it can't change prices on tickets its already sold.
  - The same applies if you think about the routes it has decided to fly.
- So solution
  - Go long in the jet fuel market
  - In fact, buy futures at the same rate as you are selling tickets.
- Makes sense because you know that you will be short in a very predictable way.

Why options?

- Now consider another of the airline's decisions.
- If business is good it wants to expand, if business is bad it will not want to do so.
  - Expansion means buying or leasing more planes
- Its business is likely correlated with that of other airlines, so when business is good the price of planes is high (or delivery times long)
  - It does not want to enter into forward contracts because it does not want the planes if business is bad.
  - What it wants is an option.

Beyond airlines

- Futures and forwards
  - Solve price uncertainty when you know you will want to go long or short in the future
- Options
  - Allow you to target specific probabilities (events)
  - Because you can combine puts and calls.
  - They also affect incentives for the holders

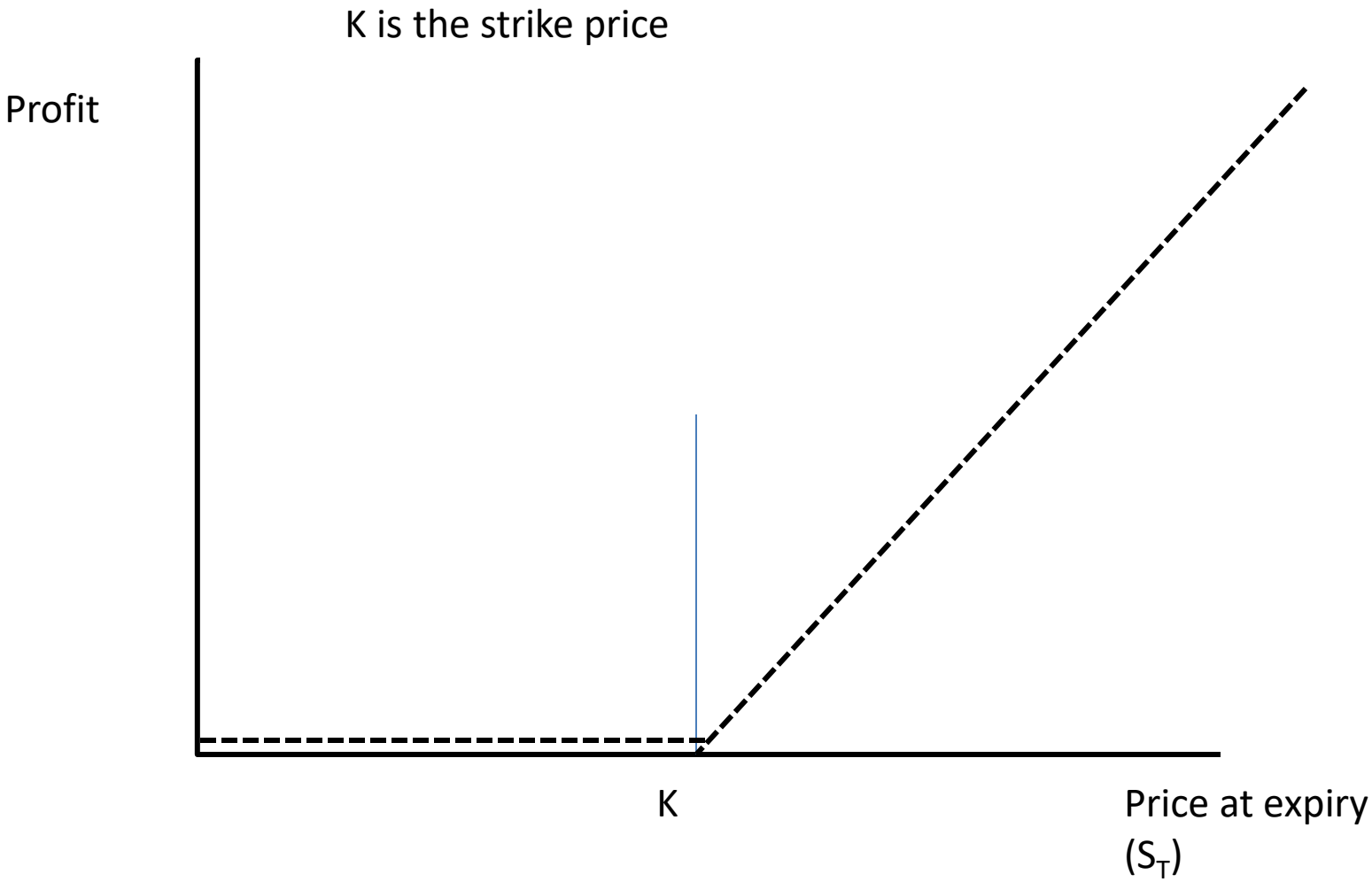
NBC-Universal

- In December 3, 2009, Comcast buys a stake in NBC Universal from GE.
- Under the agreement, NBC Universal is 51% owned by Comcast and 49% by GE. Comcast paid $6.5 billion cash to GE.
- Comcast reserves the right to buy out GE's share at certain times
- GE reserves the right to force the sale of their stake within the first seven years.
- February 12, 2013 announcement that Comcast intends to complete the $16.7 billion purchase early, all at once.

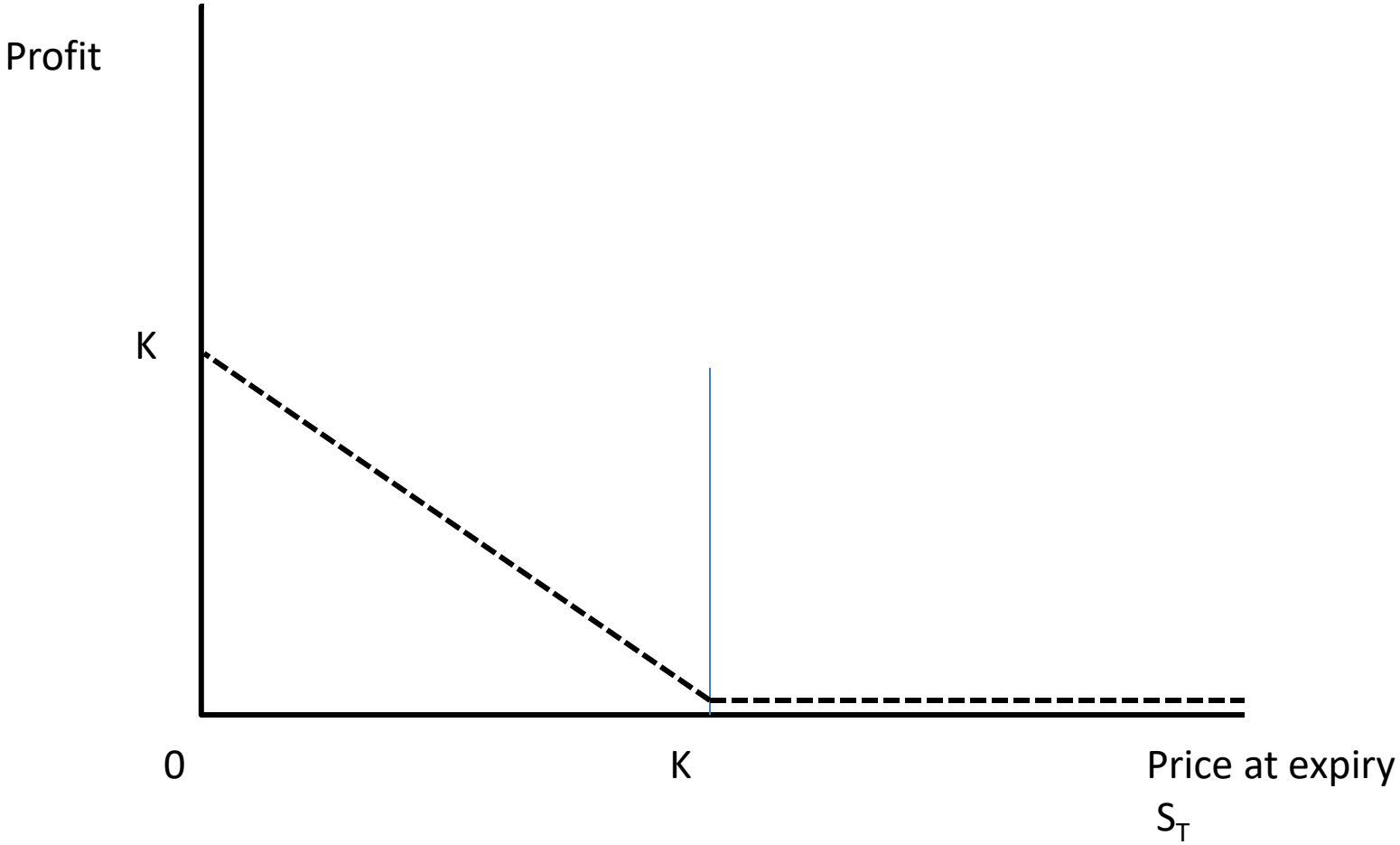
Why both reserved rights

- The merger is supposed to create synergies
  - Both sides stand to gain, but its risky
- On the GE side
  - GE is handing over control, wants to ability to get out if do not like what is happening.
  - They want to induce Comcast to make the venture profitable
- On the Comcast side
  - They want control
  - They want a large part of the upside
  - But they want protection.
- If Comcast does a bad job has to buy GE's remaining stake (expensive)
- If it does a good job then it wants the whole firm and does not want GE to be able to hold things up. So find mechanisms to split profits.

Call option



Put option



Another zero sum game

- Leaving aside the transaction technology
- Put and call options at the same strike price are never both exercised
- The counter party's return to a call option are essentially the losses the holder of a put option avoids by not exercising his option.
- The holder of an option can never make a loss (after buying it). So the counter party can make a profit only if the option is not exercises.
- So options must have a positive initial price
  - To compensate the counter party for the expected loss
  - That price is going to be different from either the futures price or the spot price of the asset.

Value of options

- Gross payoff is the difference between the strike price (K) and the price at expiry (S_T)
- If $K<S_T$ then call option (but not put) valuable
- if $K>S_T$ then put is valuable (but not Call)
- Net Pay off has to include the price at which you buy the option (C), and the income foregone
- For a call option
  - $\Pi(K)=(S_T-K)-C(1+r)^T$
- For a put option
  - $\Pi(K)=(K-S_T)-C(1+r)^T$

Equity claims as option (and leverage)

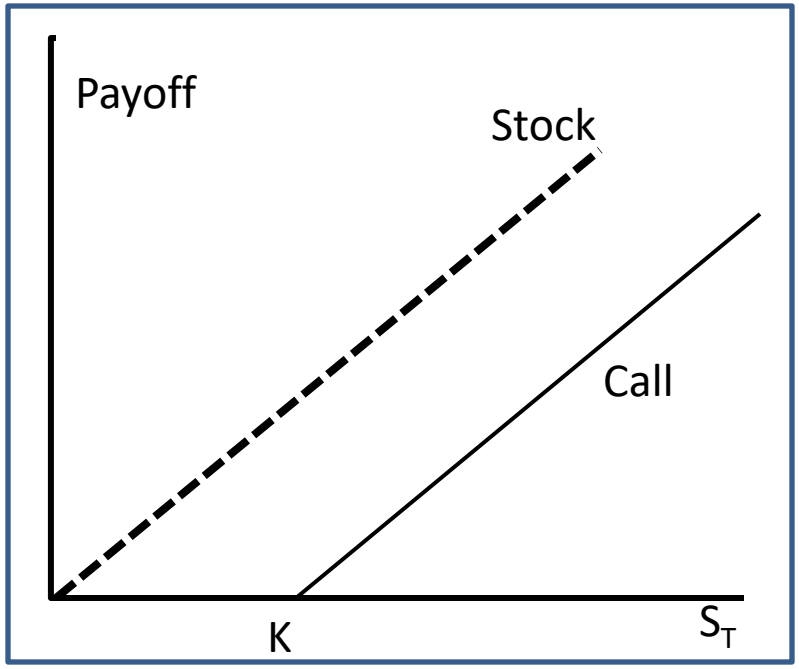
- Firm has $50 in assets
- Its liabilities are $25 in equity and $25 in bonds.
- The bonds have face value of $50.
- So now consider the payoffs next period

Firm value	Payments to bonds	Payments to Equity	Value of a call on the firm at 50
20	20	0	0
30	30	0	0
40	40	0	0
50	50	0	0
60	50	10	10
70	50	20	20
80	50	30	30
90	50	40	40

Option Prices

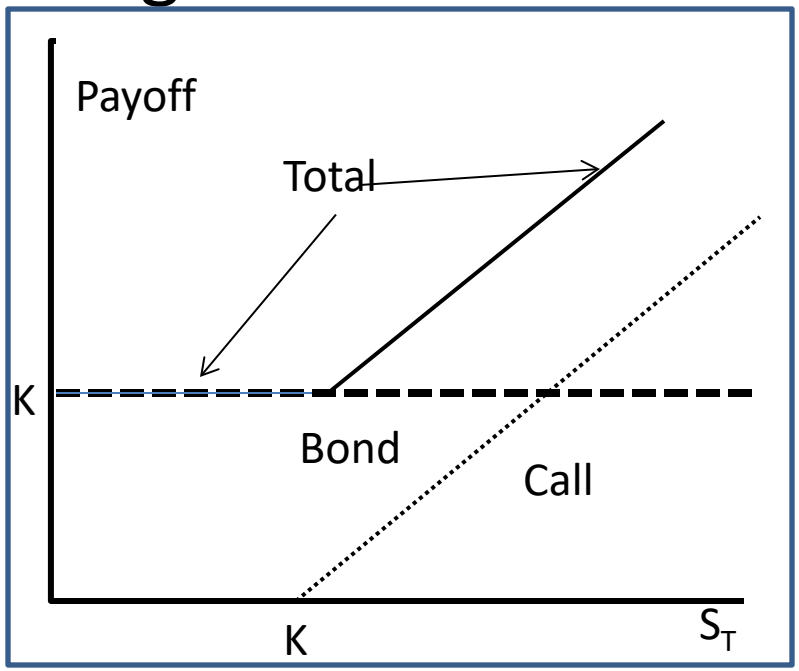
- The bounds approach (assume K=S, strike price is current price)
- C>0 (the option is not free)

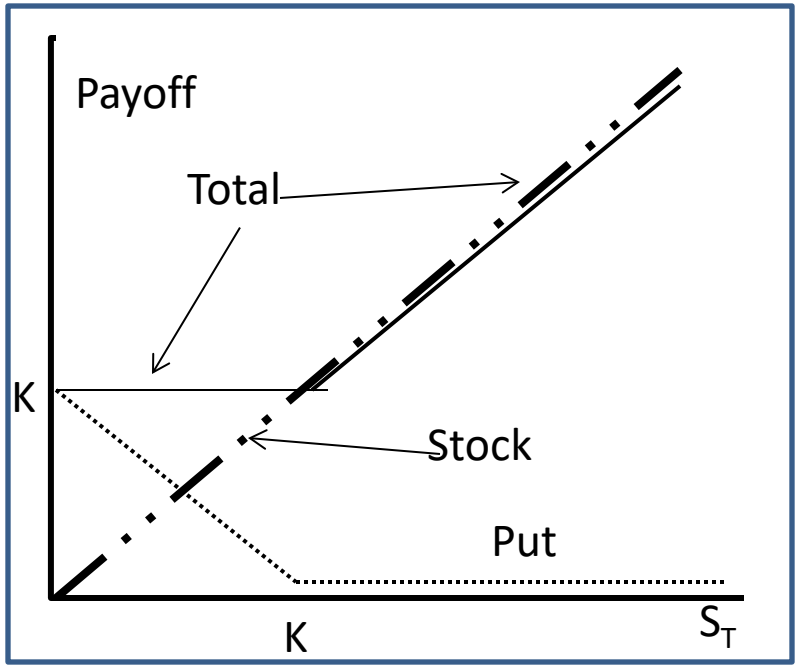
- C<S
  - Could buy stock or buy call at strike price K
  - At expiry call earns $K-S_T$ if (S_T>K), 0 otherwise
  - At expiry stock is worth S_T



Put Call parity

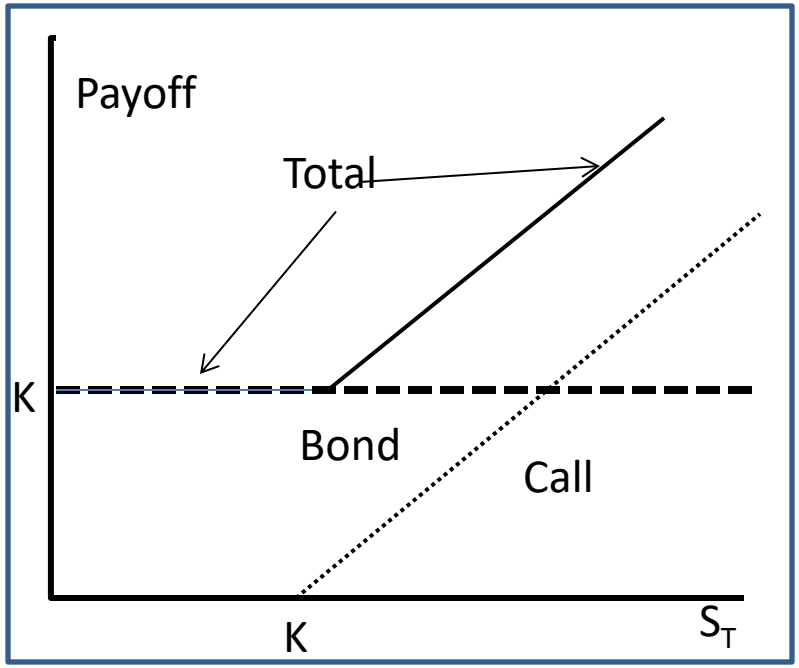
given K Call+bond is same as put +stock





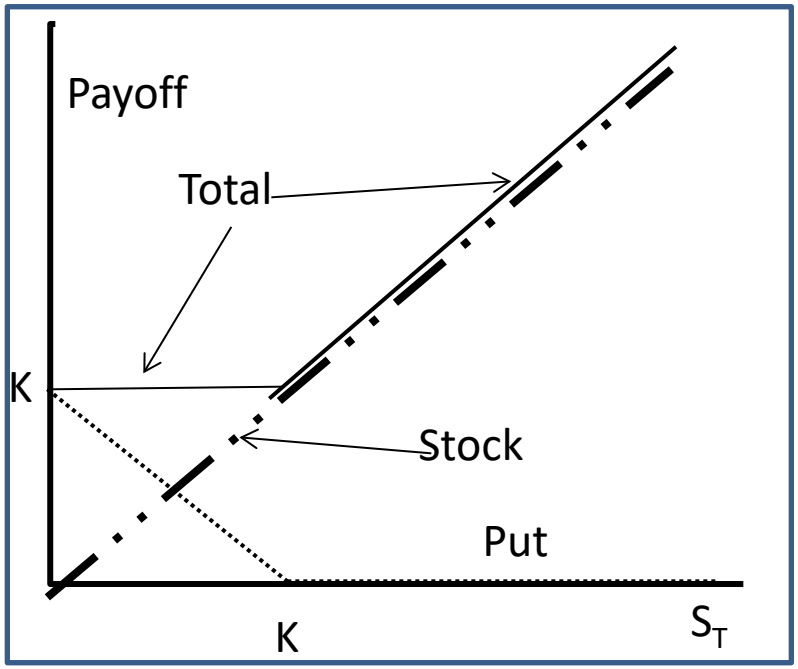
- Here we have a call with strike price of K and bond with face value of K
- Here we have a put with strike price of K and one share of stock (S=k)

Put Call parity Step 1
Value K Call+bond



- At expiry
- If $S_T \leq K$
- The call option has no value so you get B
- If $S_T > K$. The option has value and you get
- $B + Call = B + S_T - K$
- But $B = K$ => $B + Call = S_T$

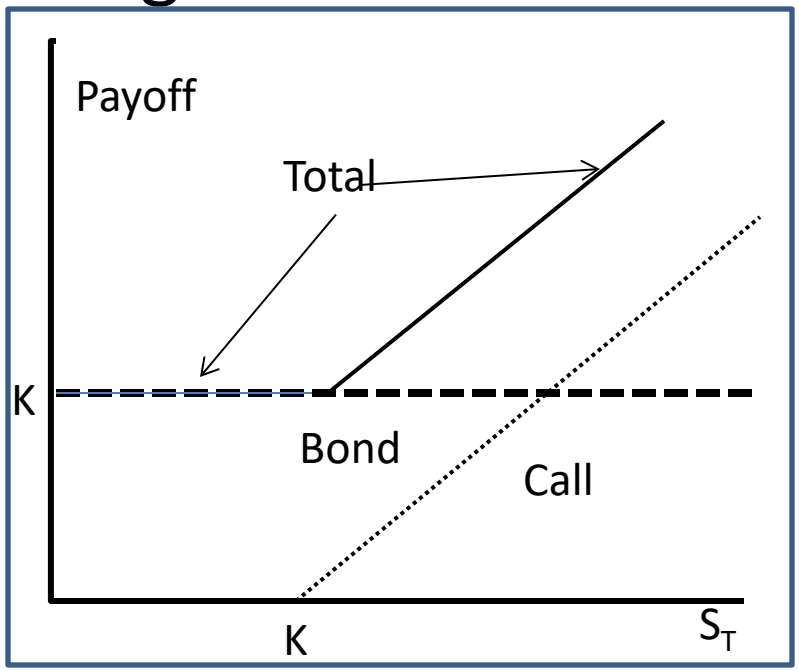
Put Call parity Step 2

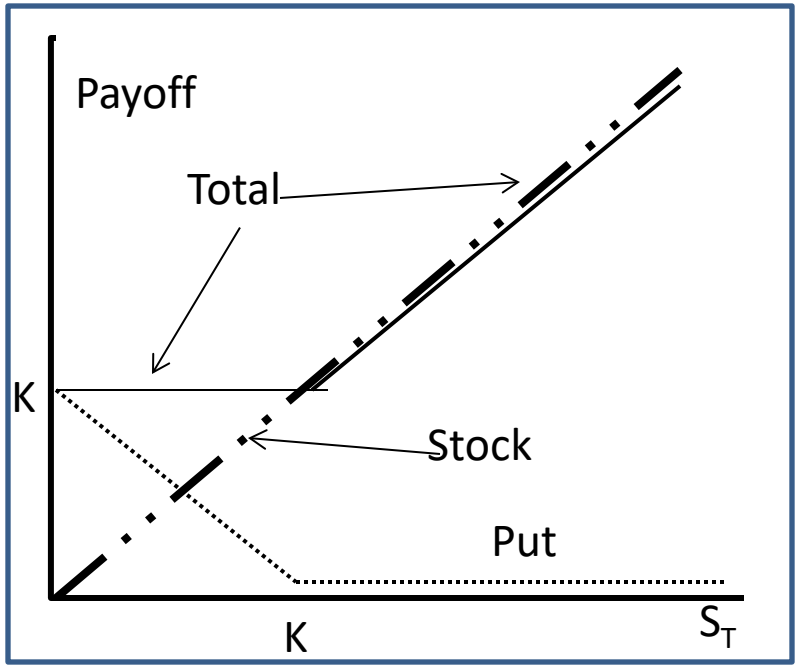


- At expiry
- If $S_T \geq K$
- the put option has no value so you get the stock
- $S_T < K$. The option has value and the stock has value
- $S_T + Put = S_T + K - S_T = K$

Put Call parity

given K Call+bond is same as put +stock





If $S_T \leq K, B + Call = K$

If $S_T > K, B + Call = S_T$

If $S_T \geq K, S_T + Put = S_T$

If $S_T < K, S_T + Put = K$

What is the price of the option? One period binomial model

Notation

- Current stock price S_o
  - With probability (q) $S_1 = S_o + \Delta$
  - With probability (1-q) $S_1 = S_o - \Delta$
- Call option has strike price $K = S_o$
  - With probability (q) $C_1 = \Delta$
  - With probability (1-q) $C_1 = 0$
- Bond has return r.
- Can invest either in stock or bond or in the call

From put call parity to replicating portfolio

- We can actually use bonds and stocks to produce a portfolio that replicates the option
- By arbitrage argument that mean that must be the price of the option.
- Consider a stock that has a current price of S_o between now and the next period it can go up by or down by Δ, the riskless rate of return is r
- Want to determine the value of an option at strike price S_o.

Replicating portfolio

- Consider a portfolio of a stock and b bonds
- Such that

1. $a(S_o+\Delta)+b(1+r)= \Delta$
2. $a(S_o-\Delta)+b(1+r)= 0$

- 1-2 => $a(S_o+\Delta)- a(S_o-\Delta)= \Delta \Leftrightarrow a2\ \Delta= \Delta$
  - $a= 1/2$
- From (2) $b=-0.5\,\frac{S_o-\Delta}{(1+r)}$
- a is the number of shares needed to replicate one call option. Also known as the option delta
  - Two state, symmetric case a=0.5

Replicating portfolio

a= 1/2

- a, the option delta does not depend on probabilities (q) but on expected values of the upside and the downside.
- Arbitrage says the cost of the replication portfolio has to be the price of the option

$$PV = aS_o + b$$

$$a = \tfrac{1}{2} \text{ and } b = -\frac{S_o - \Delta}{2(1+r)}$$

$$PV = 0.5S_o - 0.5\frac{S_o - \Delta}{1+r} \text{ or } PV = 0.5\frac{(1+r)S_o - S_o + \Delta}{1+r}$$

$$PV = \frac{rS_o + \Delta}{2(1+r)}$$

- There is also no q in here.
- ARBITRAGE (again)

Example

- Current stock price S_o =55, Δ=20
- Call option has strike price K= 55
- Bond has return r=0.05
- a= ½
- Cost of half a share (aS_o)= 27.5
- b=-a(S_o-Δ)/(1+r)=-0.5(55-20)/1.05
  =-17.5/1.05=-16.16
- Cost of Bond b= -16.16
- Cost of portfolio=27.5-16.16=11.34
- We have just priced a binomial option

Beyond the Binomial

- Can show that as Δ gets small and time gets short
- The evolution of price changes approximated a normal distribution
- This implies that the option price is a PDE
- Black Scholes solved this pb
- Take the option's class

Recap

- Options have value.
- Many options are imbedded in contracts
  - Collateral, margin accounts…
- Pricing options depends
  - Call option expected value above the strike price
  - Put option expected value below the strike price
- Price via replicating portfolio